RythuMitra
A Multilingual Voice-First AI Farming Companion

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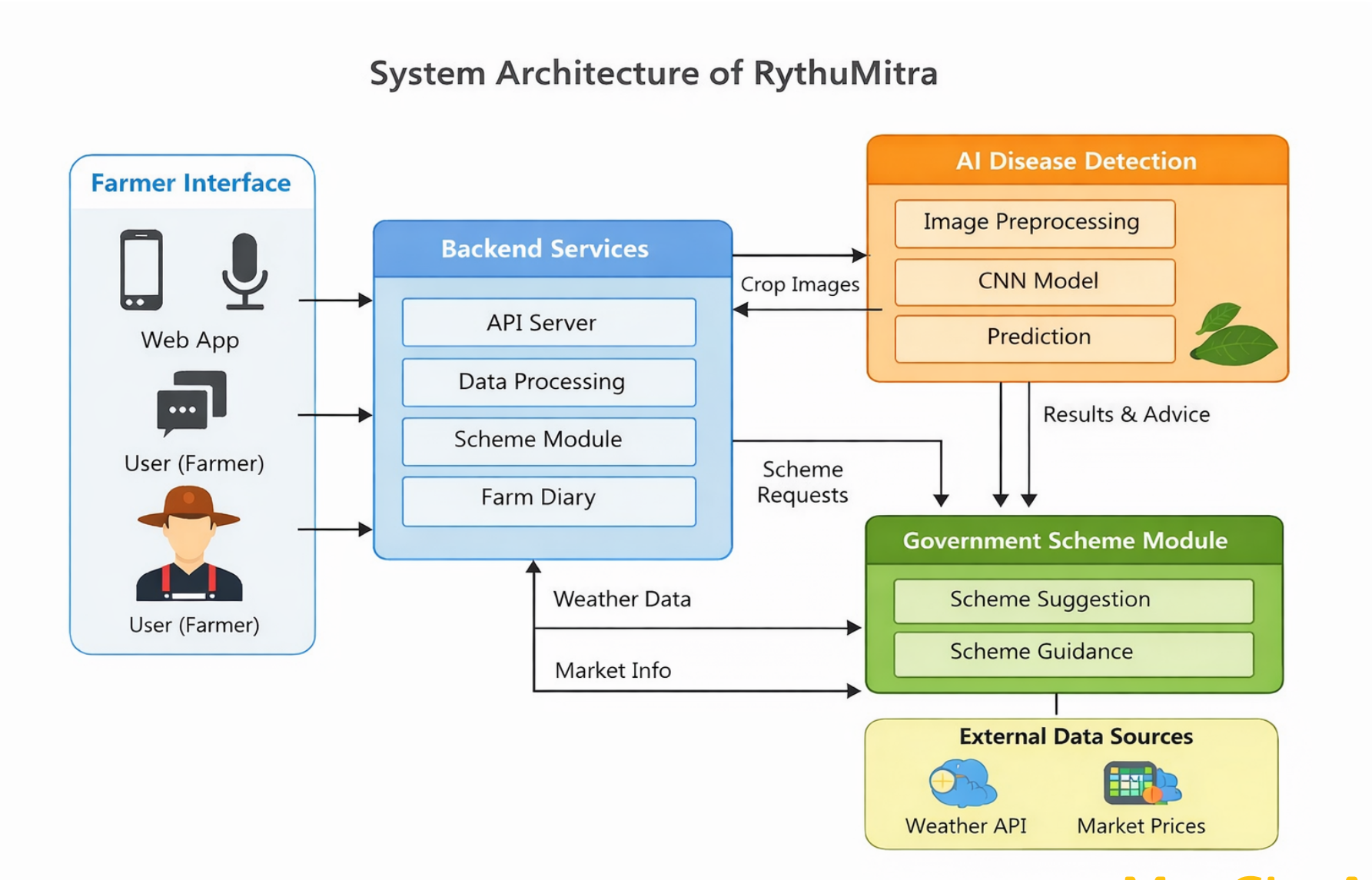
Overview

1. Summary of Stage I
2. Implementation
3. Results & Outcomes

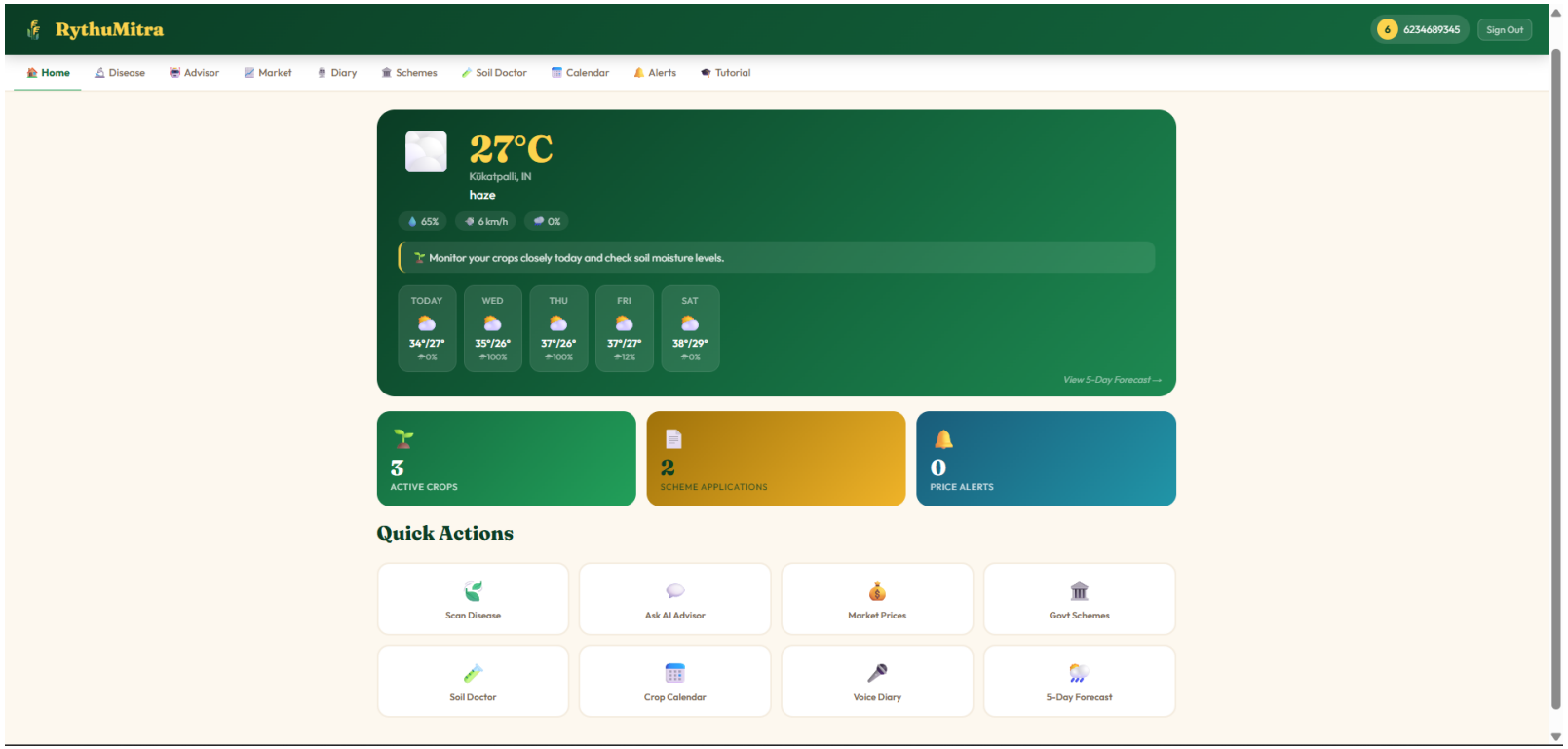
Summary of Stage I

- Farmers face significant challenges in early crop disease detection, transparent access to government schemes, and timely market and weather information.
- Existing digital solutions are fragmented and do not provide an integrated farmer-centric ecosystem.
- **Objective:** To design and develop a unified Farmer Super App integrating Artificial Intelligence and Blockchain technologies for smart agricultural support.
- Implemented real-time image-based crop disease detection using CNN models with personalized fertilizer and pest-control recommendations.
- Developed a blockchain-enabled "Easy Apply" feature using Solidity smart contracts and IPFS for secure and tamper-proof document handling.
- Integrated additional modules including irrigation guidance, voice-driven farm diary, and conversational market & weather alerts.

Implementation – System Architecture

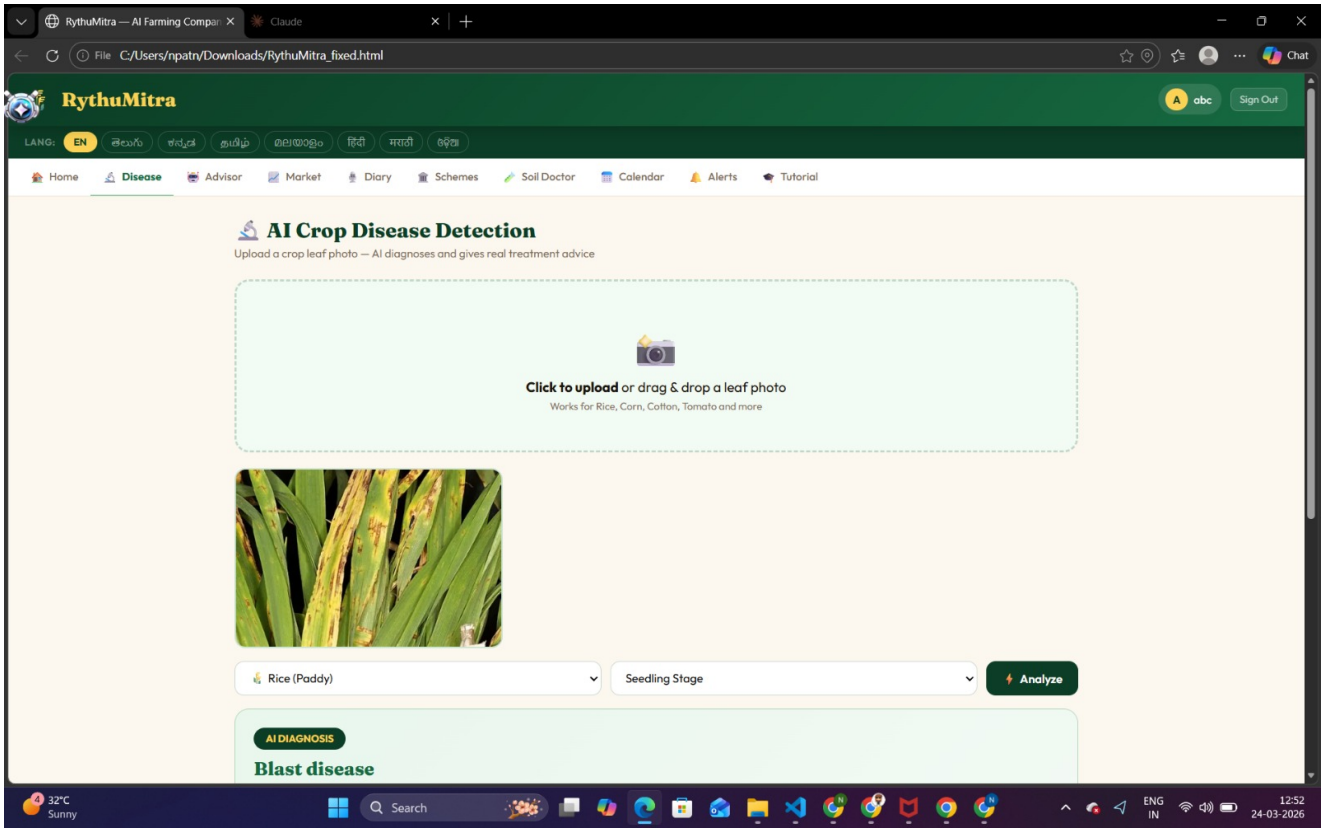


Implementation – Home Dashboard



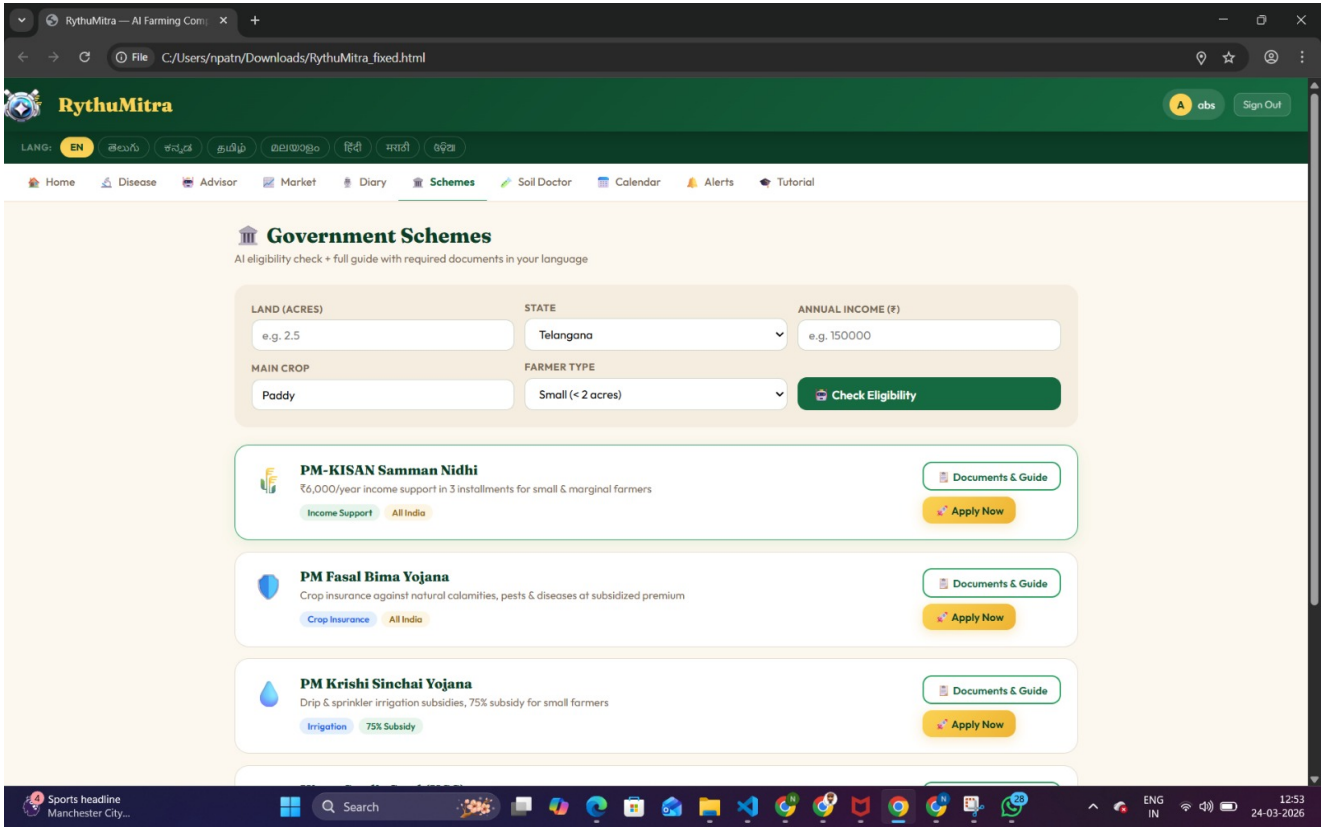
RythuMitra Home Dashboard – Real-time weather, crop statistics, and Quick Action navigation tiles.

Implementation – AI Crop Disease Detection



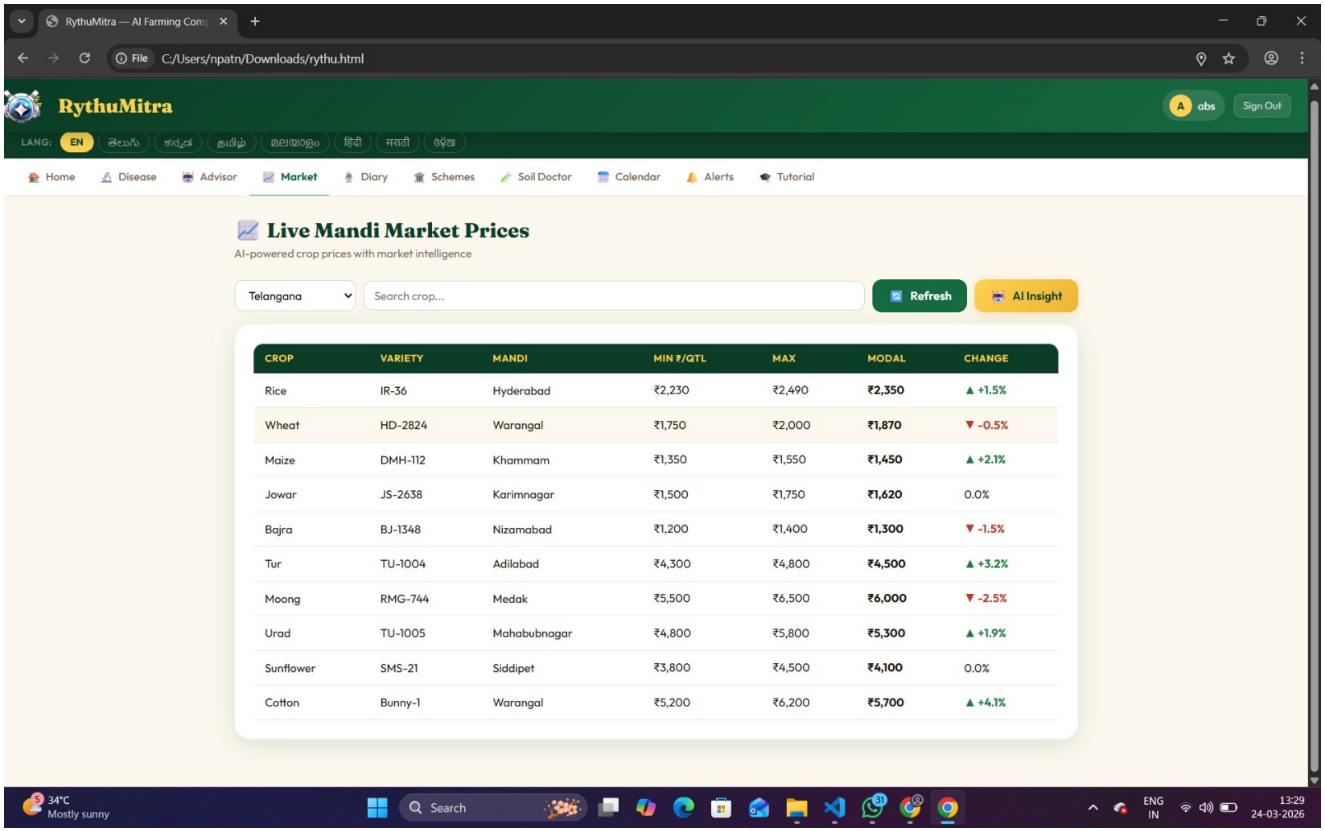
RythuMitra – AI Crop Disease Detection Module – The farmer uploads a crop leaf photo which is analyzed by CNN model to provide real-time disease diagnosis with treatment recommendations.

Implementation – Government Schemes Eligibility



RythuMitra – Government Schemes Module – The farmer enters land size, state, annual income, crop, and farmer type which is analyzed by the AI eligibility engine to instantly match and display relevant government schemes with document guides and direct application links.

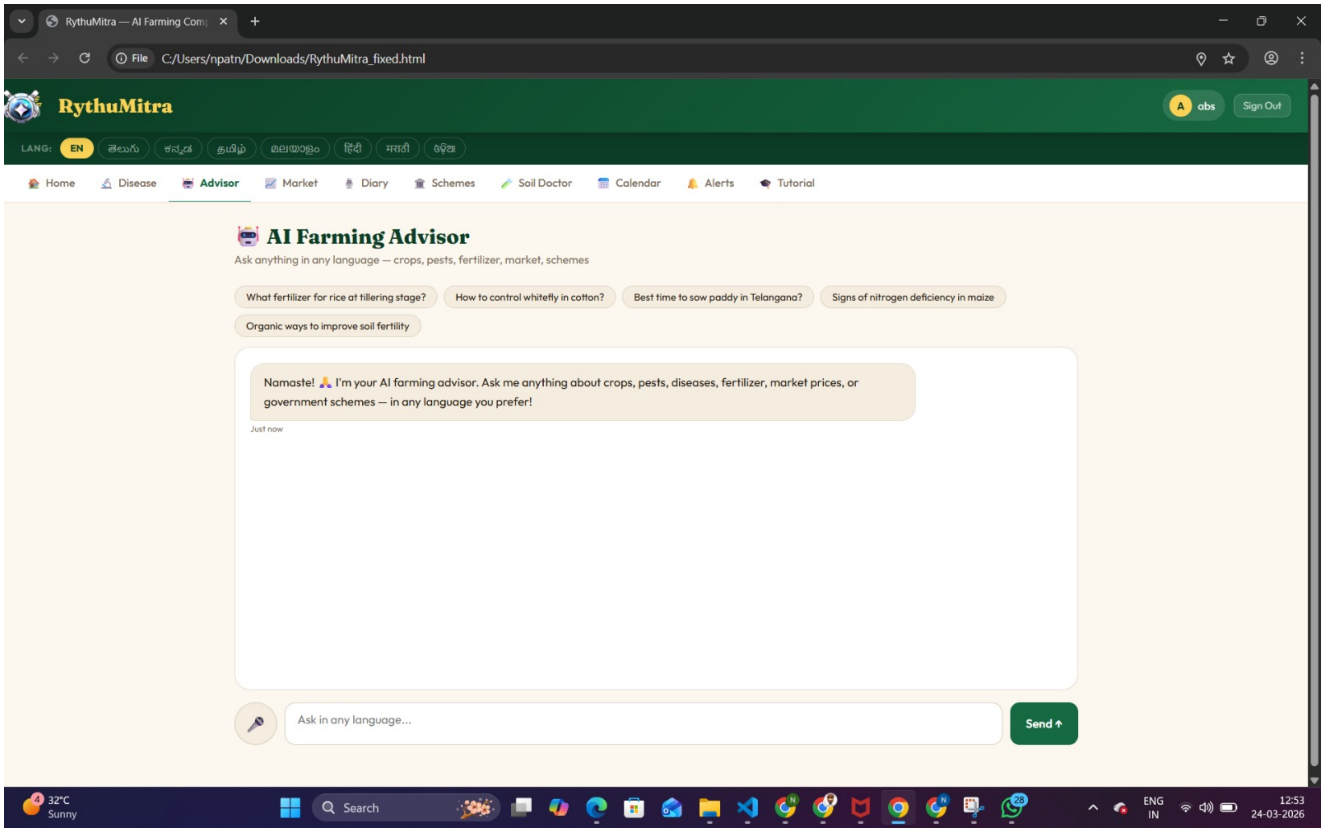
Implementation – Live Mandi Market Prices



CROP	VARIETY	MANDI	MIN ₹/QTL	MAX	MODAL	CHANGE
Rice	IR-36	Hyderabad	₹2,230	₹2,490	₹2,350	▲ +1.5%
Wheat	HD-2824	Warangal	₹1,750	₹2,000	₹1,870	▼ -0.5%
Maize	DMH-112	Khammam	₹1,350	₹1,550	₹1,450	▲ +2.1%
Jowar	JS-2638	Karimnagar	₹1,500	₹1,750	₹1,620	0.0%
Bajra	BJ-1348	Nizamabad	₹1,200	₹1,400	₹1,300	▼ -1.5%
Tur	TU-1004	Adilabad	₹4,300	₹4,800	₹4,500	▲ +3.2%
Moong	RMG-744	Medak	₹5,500	₹6,500	₹6,000	▼ -2.5%
Urad	TU-1005	Mahabubnagar	₹4,800	₹5,800	₹5,300	▲ +1.9%
Sunflower	SMS-21	Siddipet	₹3,800	₹4,500	₹4,100	0.0%
Cotton	Bunny-1	Warangal	₹5,200	₹6,200	₹5,700	▲ +4.1%

RythuMitra – Live Mandi Market Prices Module – The farmer selects their state to view real-time crop prices across major mandis with min, max, and modal rates, powered by AI market intelligence and instant price change indicators.

Implementation – AI Farming Advisor



RythuMitra – AI Farming Advisor Module – The farmer asks any question in any language about crops, pests, fertilizer, market prices, or government schemes and receives instant AI-powered advice via a conversational chatbot interface.

Results & Key Outcomes

- Successfully detected crop diseases with high accuracy using CNN model
- AI eligibility engine matched farmers to relevant government schemes instantly
- Blockchain "Easy Apply" feature ensured tamper-proof document handling via IPFS
- Conversational AI Advisor responded in 8+ Indian regional languages
- Integrated weather, market prices, and voice alerts in a single unified platform
- Reduced farmer effort from multiple visits to a single-click application process

Conclusion & Future Scope

- **Conclusion:** RythuMitra successfully integrates AI and Blockchain into a unified, multilingual farmer-centric platform
- **Future Scope:**
  - Offline mode for rural areas
  - Push notifications for price and weather alerts
  - Instant camera scan for disease detection

Thank You!